

Actual Script

Title slide

Good morning, I'm Alex and now I'll present my final year project - smart antenna beamforming for remote sensing, more specifically, a novel optimisation for bit-level faults. I'll explain what all that means but In short, we're about to repair a broken antenna in clever ways by adjusting the control algorithm.

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I would like to acknowledge the traditional owners of the land on which I study and work - the Wurindjeri people of the Kulin nations. I pay my respects to their elders, past, present and emerging.

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- Remote sensing is the gathering of physical information from a faraway location. Such as measuring the moisture of earth's surface all the way from space. We typically do this with satellite antennas that listen to radio waves bouncing off of earth's surface. We wanna create a map, so you wanna filter where the energy is coming from, so that fill in the right spot on the map.
- The simplest dipole antenna has a pretty even radiation pattern, so whenever the antenna gets excited, we don't really know where the energy is coming from.
- Beamforming is the practice of shaping this radiation pattern to be highly concentrated in one direction, by placing an array of tiny simple antennas together.
- Beam steering is what's happening in the animation, you can change the direction of this beam by tweaking the time delays in all the radiating elements.
- This is a key enabling technology in 5G, radar, satellite communication and sensing.

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- We accomplish all this with phased array antennas - just a bunch of little antennas placed in an array.
- Notice how when we apply the appropriate time delay also called a phase delay to these elements, the small wavefronts line up to create a powerful wavefront travelling in some off-direction.
- The beamsteering and beamforming are controlled by controlling these time delays or phase delays, and the device that does this is a phase shifter.

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In this project, I was provided with an unassembled and faulty phased array antenna from a PhD student, to get it working. It had low resolution phase shifters as shown on the right. Think of them as a series of valves that determine whether a signal takes a shorter or longer path. With 4 valves each responsible for a half cycle delay, quarter cycle delay etc, this divides the possible delay phases into 4-bit levels, which is 16 levels. These are digital phase shifters, because they can't create arbitrary phase delays, but they get the job done.

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Unfortunately the hardware was very prone to damage, due to these fragile diodes that were less than a millimeter in size. Every failed diode takes out one of the bits in the phase shifter. Which would damage the radiation pattern severely.

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Now is a good time to mention that the unrecoverable fault states of the antenna provided major challenges and setbacks, seen in that big gap in the middle. But I managed the project by pivoting and narrowing the scope to something both achievable and novel.

I recovered progress with rapid simulation, restructured milestones, and completed all algorithm design and testing tasks.

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What I ended up with, was this research question:

How can the performance of a phased array antenna be affected when low-resolution digital phase shifters suffer from bit-level faults?

What algorithmic solutions can mitigate the impact of these hardware faults?

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Let's look at what fault models have been considered in literature. They fall into 4 categories.

Total element drop out; each phase shifter gets a phase kick due to hardware imperfections; entire phase shifter being stuck to some random phase and can't be controlled anymore; and finally, damage done on the radiation pattern level.

What we have is individual bit failure that results in a partially constrained phase shifter.

We have identified a novel problem.

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So we developed a mathematical model of bit-level faults. The equations on the left describe the quantisation of phase shifts due to using digital phase shifters. The equation on the right describe the altered phase due to bits being stuck to 0 or 1.

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With this, we were able to simulate how the radiation pattern gets damaged. This is what it typically looks like.

The translucent pattern is the ideal pattern, while the opaque ones are the damaged patterns.

On the left we have what happens when the faulty bits are random, and on the right we have what happens when the faulty bits happen to be in the most significant binary position. We have now seen that even with very few broken diodes, the damage to the pattern is ghastly.

The next step? Is to fix it.

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Several engineering constraints shaped our solution. First, no hardware redundancy. We assume the antenna is to be mounted onto a remote sensing satellite like this one by NASA.

And that faults occur mid-deployment. The extreme size and weight constraints of a space mission forbid placing spare phase shifters on the board.

Then there is power efficiency. Any algorithm must run quickly to avoid draining system resources.

Then, sustainability. By extending the usable life of compromised systems, we can reduce electronic waste which has a benefit both for cost and environmental impact.

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So these are the most popular radiation pattern repair solutions in literature. The genetic algorithm and particle swarm optimisation, which are global optimisation methods that both utilise a large random population that stochastically approach the optimal solution upon many many iterations. These are very time consuming and rather complex, and are often run before the mission to create these massive look-up tables of what phase settings should be set given a desired beam direction.

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We developed a comparatively very simple algorithm to solve our bit-level problem. It is an element-wise local optimisation (ELO). Here's the algorithm, and here is how it works.

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First imagine a single phase shifter that wants to output 0000 corresponding to no phase delay. And that the diodes have been burnt to short circuit, fixing the most significant bit to 1. So the phase shifter actually outputs 1000 - a half cycle phase shift, which is as far away from the original phase as possible. ELO looks at the remaining possible phases and calculates the phase setting with the least wrapped distance from the original setting, reducing the phase deviation down to only 22.5 degrees. The control code then sets those remaining 3 bits to 1 instead of 0 to the phase shifter. This is repeated for all elements in the phased array.

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This is how ELO looks like in action for a typical fault scenario. The black trace is the ideal pattern, red means damaged, and the blue is the repaired pattern. We see a significant improvement, most times to the near ideal pattern, turning a wasted antenna into something that's highly functional. And that is the main achievement of this project. We have used software to fix hardware.

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I have adapted the algorithms used in literature to this scenario which had a binary search space. And their performances are shown here. They seem to do the job equally as well.

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To obtain statistical information about their performance, I ran monte carlo trials - 1000 trials per fault configuration. Had to parallelise so that it wouldn't take a week for each run. I measured the performance with common metrics in the literature, such as mean squared error of the radiation pattern - the lower the better. For practical purposes, the proportion of bits stuck below

0.2 is the region of interest. Above this, you'd just throw the antenna away. The proposed algorithm performs better than particle swarm, and slightly worse than genetic algorithm.

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But here is the real advantage. The proposed algorithm ELO runs an average of 0.2 ms. while the global optimisations take 4-5 whole seconds, for not much improvement. This means ELO can actually run in real time as we scan the earth through all these steering angles.

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The impact of this work lies in its practicality and novelty. We developed a bit-level fault model, encountered in the lab, not found in literature. We developed an algorithm that repaired the array to near optimal performance in real time. Validated with monte carlo simulations. We showed that the simplest method was effective and the most efficient. This allows for longer-lasting, fault-aware sensing systems.

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Next, one could look into obtaining statistical failure data for phase shifter bits; embed this robust beamforming algorithm into a working hardware demonstration. And one could investigate real time fault detection methods, perhaps with machine learning, that when combined with the repair algorithm, forms a truly self-repairing antenna in real time.

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So that's all from me. Thank you for listening, and I welcome any questions from the panel.